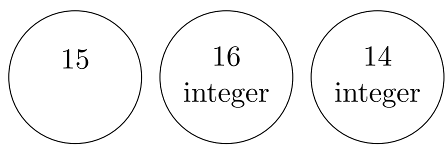
IEOR 4004 — Practice for Quiz 2

Solutions

Problem 1.

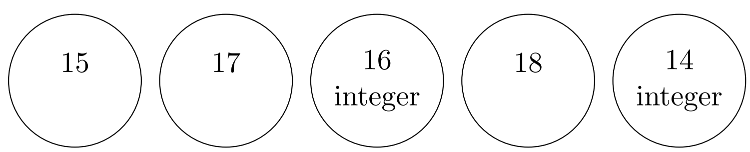
a. Since we already have a candidate maximal integer solution with objective value 16, we can prune all nodes with less than 16 objective value and itself by definition of algorithm, that is,



b. Since we can only have candidate maximal integer solutions with greater than 16 objective value from children nodes of those nodes with objective value 17 and 18, the upper bound is 18;

c. Since we already have a candidate maximal integer solution with objective value 16, the lower bound is 16;

d. a'. Similarly, for minimization problem, since we already have a candidate minimal integer solution with objective value 14, we can prune all nodes with greater than 14 objective value and itself by definition of algorithm, that is,



b'. Since we already have a candidate minimal integer solution with objective value 14, the upper bound is 14;

c'. Since there is no node with objective value less than 14, we have no way to have a candidate integer solution with less than 14 objective value, thus the lower bound is also 14. In this case, we can determine the optimal integer solution is 14 then the upper bound and the lower bound meet at 14.

Problem 2. 1. True, since every integer solution to a subproblem is an integer solution to the original problem.

2. False in general. Consider, e.g., the original suproblem. We know that the value of its LR is an upper bound to the optimal solution of the original IP – in particular, it can be strictly larger (e.g.[1] in the problem $\max x_1 : x_1 \leq 1.5, x$ integer).

[1] You are not required to give a counterexample to obtain full points for this question.

3. False in general. It may be that the optimal solution of one of the two suproblems has also value v. Consider[2] the problem

$$\max x_2 : x_2 \leq 2, x_1 + x_2 \leq 2.5, x_1, x_2 \geq 0 \text{ and integer.}$$

An optimal solution of the LR is $(.5, 2)$. Branching on x_1 leads to two suproblems. The one obtained by adding $x_1 \leq 0$ has optimal integer solution $(0, 2)$, which has same value as $(.5, 2)$.

4. True, since the set of feasible integer solutions of the original problem is the union of the sets of integer solutions of all the subproblems.

Problem 3.

a) The rightmost node, of value 16, since there is an integer solution of value 17 (Recall that the problem is in maximization form).

b) The optimal integer solution has value at least 17 (integer solution of highest value found so far) and at most 19 (solution of highest value in any node whose children have not been both explored).

c) They do not change.

Problem 4. See the end of the document.

Problem 5.

a)

$$\begin{aligned}
\max\ & 13x_1 + 11x_2 \\
\text{s.t. }\ & x_1 + x_2 \leq 10, \\
& x_1^2 + 4x_2^2 + 4x_1x_2 \leq \alpha, \\
& x_1 \geq 0, \\
& x_2 \geq 0.
\end{aligned}$$

b) The previous problem can be rewritten as

$$\begin{array}{ll}
\min & -13x_1 - 11x_2 \\
\text{s.t.} & x_1 + x_2 \leq 10, \\
& x_1^2 + 4x_2^2 + 4x_1x_2 \leq \alpha, \\
& x_1 \geq 0, \\
& x_2 \geq 0
\end{array} \tag{$\circ$}$$

Since the objective function is convex, it suffices to show that $(\circ)$ is a convex set. We do that in three steps

i) We know that

$$C_0 = \left\{ (x_1, x_2) \;\middle|\; \begin{array}{l} x_1 + x_2 \leq 10, \\ x_1 \geq 0, \\ x_2 \geq 0 \end{array} \right\}$$

is a convex set.

ii) We next show that

$$C_1 = \{x \mid x_1^2 + 4x_2^2 + 4x_1x_2 \leq \alpha\}$$

[2] You are not required to give a counterexample to obtain full points for this question.

is a convex set, where

$$g(x) = x_1^2 + 4x_2^2 + 4x_1x_2.$$

Indeed,

1. g is a convex function, since it is twice differentiable and

$$\frac{\partial^2 g}{\partial x_1^2} = 2 \geq 0, \qquad \frac{\partial^2 g}{\partial x_1^2}\frac{\partial^2 g}{\partial x_2^2} - \left(\frac{\partial^2 g}{\partial x_1 \partial x_2}\right)^2 = 16 - 16 = 0 \geq 0.$$

2. We know by a previous exercise that the set $\{x : f(x) \leq \alpha\}$ is convex for f convex.

iii) Last, observe that the intersection of two convex sets is convex. In particular, $C_0 \cap C_1$ is convex. We deduce that the feasible region of the problem is convex.

c) The vector of expected returns is

$$\mu = \begin{pmatrix} 13 \\ 11 \end{pmatrix}.$$

The variance of the portfolio is $x_1^2 + 4x_2^2 + 4x_1x_2$. We know from class that the problem

$$\begin{array}{ll} \min & z^T Q z \\ \text{subject to} & \mu^T z = 1, \\ & \sum_{i=1}^n z_i \leq By, \\ & z \geq 0, \\ & y \geq 0 \end{array}$$

is convex. From its optimal solution (z^*, y^*), we can get the portfolio maximizing the Sharpe ratio as

$$x^* = \frac{z^*}{y^*}.$$

Problem 6. See the end of the document.

Problem 7. 1.

$$\begin{array}{llllllll} \min & x_u & +x_w & +x_v & +x_p & +x_q & \\ & x_u & +x_w & +x_v & +x_p & +x_q & \geq 1 \\ & x_u & +x_w & & & & \geq 1 \\ & x_u & & +x_v & +x_p & & \geq 1 \\ & x_u & & +x_v & +x_p & +x_q & \geq 1 \\ & x_u & & & +x_p & +x_q & \geq 1 \\ & x & & & & & \in \{0,1\}^5 \end{array}$$

2.

$$\begin{array}{ll} \min & \sum_{u \in V} x_u \\ & \sum_{u \in \{v\} \cup \delta(v)} x_u \geq 1 \quad \text{for all } v \in V \\ & x \in \{0,1\}^V \end{array}$$

where $\delta(v) = \{u \in V : uv \in E\}$.

3. Let S be a vertex cover and $v \in V$. Consider any $uv \in E$. Then, since S is a vertex cover, either $u \in S$, or $v \in S$. Hence, S contains either v or one of the node adjacent to v. Repeating the argument for all v we observe that S is a dominating set.

Problem 8.

a). As seen in class, the recursive formula is

$$f(i,j) = \begin{cases} 0, & i = 0, \\ \max\limits_{k\in\{0,1,\ldots,j\}} \{f(i-1, j-k) + c_i(k)\}, & i \geq 1, \end{cases} \tag{1}$$

We can apply it to fill the table as follows (here we only report the final result; we do computation more explicitly in the next exercise, which is very similar):

$$\begin{aligned} f(2,2) &= 8; \\ f(2,4) &= 14; \\ f(3,3) &= 13; \\ f(3,4) &= 20. \end{aligned}$$

b) The value of the optimal solution can be read in $f(4,4)$ and it is equal to 20. Since $20 = f(2,0) + c_3(3)$, it corresponds to investing \$4 in investment 3.

c) $f(i,j)$ is now define for $i = 0,1,2,3$ and $j = 0, .25, .5, .75, 1, 1.25, \ldots, 3.75, 4$. The recursion stays the same:

$$f(i,j) = \begin{cases} 0, & i = 0, \\ \max\limits_{k\in\{0,1,\ldots,j\}} \{f(i-1, j-k) + c_i(k)\}, & i \geq 1, \end{cases} \tag{1}$$

Problem 9. a). As seen in class, the recursive formula is

$$f(i,j) = \begin{cases} 0, & i = 0, \\ \max\limits_{k\in\{0,1,\ldots,j\}} \{f(i-1, j-k) + c_i(k)\}, & i \geq 1, \end{cases} \tag{1}$$

We can apply it to fill the table as follows

$$f(2,2) = \max\{f(1,2) + c_2(0),\ f(1,1) + c_2(1),\ f(1,0) + c_2(2)\} = \max\{7,8,6\} = 8;$$

$$\begin{aligned} f(2,4) &= \max\{f(1,4) + c_2(0),\ f(1,3) + c_2(1),\ f(1,2) + c_2(2),\ f(1,1) + c_2(3),\ f(1,0) + c_2(4)\} \\ &= \max\{11,12,13,14,12\} = 14; \end{aligned}$$

$$f(3,3) = \max\{f(2,3) + c_3(0),\ f(2,2) + c_3(1),\ f(2,1) + c_3(2),\ f(2,0) + c_3(3)\} = \max\{11,13,13,12\} = 13;$$

$$\begin{aligned} f(3,4) &= \max\{f(3,4) + c_3(0),\ f(3,3) + c_3(1),\ f(3,2) + c_3(2),\ f(3,1) + c_3(3),\ f(3,0) + c_3(4)\} \\ &= \max\{14,16,16,17,13\} = 17. \end{aligned}$$

	0	1	2	3	4
0	0	0	0	0	0
1	0	5	7	9	11
2	0	5	**8**	11	14
3	0	5	10	**13**	**17**

b) The value of the optimal solution can be read in entry $(3,4)$ of the table, and it is equal to 17. From the computation of $f(3,4)$, we see that the maximum in (1) is realized in $f(2,1) + c_3(3)$. Thus, in the optimal solution, we invest 3 dollars in investment 3. Since

$$f(2,1) = \max\{f(1,1) + c_1(0),\ f(1,0) + c_1(1)\} = \max\{5,3\} = 5,$$

we invest 0 dollar in investment 2. Thus, we invest the remaining dollar in investment 1. In formulas, the optimal solution is

$$(x_1^*, x_2^*, x_3^*) = (1, 0, 3).$$

c) Since the optimal solution already invests 0 dollars in investment 2 and the profit of investment 2 has decreased, the new optimal solution will still invest 0 dollars in investment 2. As the returns of the other investments have not changed, the optimal solution does not change.

Problem 10. (Notice that this is a convex optimization problem, not a dynamic programming problem)

a)

$$\begin{array}{ll} \max & 11x_1 + 14x_2 - (4x_1^2 - 9x_2^2 - 6x_1x_2 - 1) \\ & x_1 + x_2 \le 10 \\ & x_1, x_2 \ge 10. \end{array}$$

The constrains define the feasible region of a linear program, hence they are convex. Let g be the objective function. Note that the problem above is equivalent to minimizing $f = -g$ subject to the constraints above. We compute

$$\frac{\partial^2 f}{\partial x_1^2} = 8 \ge 0, \qquad \frac{\partial^2 f}{\partial x_1^2}\frac{\partial^2 f}{\partial x_2^2} - \left(\frac{\partial^2 f}{\partial x_1 \partial x_2}\right)^2 = (8) * (18) + 36 \ge 0.$$

We conclude that f is convex, and the problem can be formulated as a convex optimization problem.

b) Change the objective function to

$$\min 4x_1^2 + 9x_2^2 + 6x_1x_2 + 1$$

and add the constraint $11x_1 + 14x_2 \ge \alpha$. The feasible region is still that of an LP - hence convex. The derivatives of the objective function have not changed – the objective function is still convex. We conclude that the problem is one of convex optimization.

c) The required constraint is

$$4x_1^2 + 9x_2^2 + 6x_1x_2 + 1 + \|x\|_2 \le \gamma. \tag{1}$$

We showed that the variance is a convex function and, in a previous exercises, that $\|\cdot\|$ is also convex. We also showed in a previous exercise that the sum of convex functions is convex. Thus, the left-hand side of (1) is a convex function. We also showed a set of the for $\{x : h(x) \le \gamma\}$ is convex when γ is a constant and h is a convex function. Thus, (1) defines a convex set. Since the intersection of convex sets is convex (also observed in a previous exercises), we deduce that the problem is still a convex optimization problem.

Problem 11.

a) Number the row, from top to bottom, as $1, 2, 3, 4$, and the columns, from left to right, as $1, 2, \ldots, 7$. For $i = 1, 2, 3, 4$, $j = 1, 2, \ldots, 7$, we want $f_i(j)$ be equal to 1 if the robot can get to cell (i, j) without passing through a trap, and 0 otherwise. Let T be the set of cells where there is a trap and note that $(1, 1) \notin T$, since $(1, 1)$ is the starting point. We have

$$f_i(j) = \begin{cases} 1 & \text{if } i = j = 1 \\ 0 & \text{else if } (i, j) \in T \\ f_i(j-1) & \text{else if } i = 1 \\ f_{i-1}(j) & \text{else if } j = 1 \\ \max\{f_{i-1}(j), f_i(j-1)\} & \text{otherwise.} \end{cases}$$

Let us explain the various cases above:

1. There is no trap in position $(1, 1)$;

2. There are traps in all cells from T;

3. If the robot is in the first column, it can get to a cell without a trap if and only if it can get to the cell immediately above it without a trap;

4. If the robot is in the first row, it can get to a cell without a trap if and only if it can get to the cell immediately to the left of it without a trap;

5. Else, the robot can get to a cell without a trap if and only if it can get to the cell immediately above or immediately to the left without a trap.

Whether the robot can get to the final cell without a trap can be read in position $f(4, 7)$.

b) Let $M = 10$. We know that, if the robot can get to a cell without a trap, it can do so in at most $6 + 3 = 9$ moves. Thus, we are going to let $f(i, j)$ to denotes the smallest number of moves the robot needs to get to cell (i, j) without a trap, and use any number $\geq M$ to denote "no path without a trap is possible". We let $f_i(j) = 0$ for $i < 0$ or $j < 0$ and write the following recursion ;

$$f_i(j) = \begin{cases} 0 & \text{if } i = j = 1 \\ M & \text{else if } (i, j) \in T \\ \min\{f_i(j-1), f_i(j-2)\} + 1 & \text{else if } i = 1 \\ \min\{f_{i-1}(j), f_{i-2}(j)\} + 1 & \text{else if } j = 1 \\ \min\{f_{i-1}(j), f_i(j-1), f_{i-2}(j), f_i(j-2)\} + 1 & \text{otherwise.} \end{cases}$$

Let us explain the various cases above:

1. There is no trap in position $(1, 1)$, and the robot starts from there;

2. There are traps in all cells from T, so the robot cannot get there;

3. If the robot is in the first column, it can get to a cell without a trap if and only if it can get to the cell immediately above, or to the one above that, without a trap;

4. If the robot is in the first row, it can get to a cell without a trap if and only if it can get to the cell immediately to the left of it, or the one to the left of it, without a trap;

5. Else, the robot can get to a cell without a trap if and only if it can get to the cell immediately above, or the one above that, or immediately to the left, of the one to the left of that, without a trap.

If the entry in $f(4, 7)$ is at most $M - 1$, that number denotes the minimum number of steps the robot needs to get to position $(4, 7)$. Else, the robot cannot get to position $(4, 7)$ without passing through a trap.

Bonus question #1. Similarly to the problem seen in class, we can formulate the problem as a dynamic program as follows:

- **stages:** at stage i we schedule first i jobs $j_1, \ldots, j_i$;

- **states:** a state is a triple (t_1, t_2, t_3) denoting that we used up to time t_1 on machine #1, up to time t_2 on machine #2, and up to time t_3 on machine #3;
- **function:**

$$f(i; t_1, t_2, t_3) = \begin{cases} 1, & \text{if it is possible to schedule the first } i \text{ jobs on the three machines} \\ & \text{using up to time } t_1, t_2, t_3 \text{ on the three machines (in this order);} \\ 0, & \text{otherwise.} \end{cases}$$

To compute $f(i; t_1, t_2, t_3)$, we either execute job j_i on machine #1, or on machine #2, or on machine #3, and we pick the option that is achievable (if any). In formulas,

$$f(i, t_1, t_2, t_3) = \begin{cases} 0, & t_1 < 0,\ t_2 < 0 \text{ or } t_3 < 0, \\ 1, & i = 0, \\ \max\{f(i-1; t_1 - p_i, t_2, t_3),\ f(i-1; t_1, t_2 - p_i, t_3),\ f(i-1; t_1, t_2, t_3 - p_i)\}, & i \geq 1. \end{cases}$$

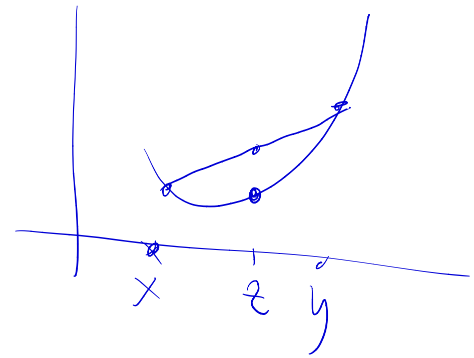
We let t_1, t_2, and t_3 go from 1 to $\sum_{i=1}^n p_i$, since the completion time cannot exceed $\sum_{i=1}^n p_i$. The optimal solution is given by the smallest t^* such that $f(n; t^*, t^*, t^*) = 1$.

Bonus question #2. An item is in some optimal solution of the knapsack problem if and only if, when its profit is increased by 1, the value of the optimal solution to the knapsack problem increases. Thus, we can do the following:

1. Let $S^* = \emptyset$
2. Solve the original knapsack problem, and let v its objective function value;
3. For $i = 1, \ldots, n$:
   (a) Solve the knapsack problem obtained from the original one by increasing the profit of item i by 1, and let v_i be its objective function value;
   (b) If $v_i > v$, add i to S^*

The required set is S^*.

a) Let $f, g : \mathbb{R}^d \to \mathbb{R}$ be convex functions. Prove that $f + g$ is a convex function.



Recall the definition of convexity

f convex if for every x, y, for every $\lambda \in [0,1]$

$$f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$$

$h = f + g$. To show:

$$h(\lambda x + (1-\lambda) y) \leq \lambda h(x) + (1-\lambda) h(y) \quad \forall x, y, \lambda \in [0,1]$$

We have:

$$h(\lambda x + (1-\lambda) y) = f(\lambda x + (1-\lambda) y) + g(\lambda x + (1-\lambda) y) \quad \text{[BY DEFIN. OF } h\text{]}$$

$$\leq \lambda f(x) + (1-\lambda) f(y) + \lambda g(x) + (1-\lambda) g(y) \quad \text{[BY CONVEXITY OF } f, g\text{]}$$

$$= \lambda [f(x) + g(x)] + (1-\lambda)(f(y) + g(y)) \quad \text{[REARRANGING]}$$

$$= \lambda h(x) + (1-\lambda) h(y) \quad \square \quad \text{[BY DEFINITION OF } h\text{]}$$

b) Prove that $\|x\|$ is convex, where $\|x\|$ is any norm. A norm $\|x\|$ of a vector x is an abstraction of the classical notion of length that satisfy the following properties.

- Triangular inequality: $\|x + y\| \leq \|x\| + \|y\|$
- Homogeneity: For $\lambda \in \mathbb{R}_{\geq 0}$, we have $\|\lambda x\| = \lambda \cdot \|x\|$
- Positiviness: If $\|x\| = 0$, then $x = 0$.

for all $x, y, \lambda \in [0,1]$

$$\|\lambda x + (1-\lambda) y\| \overset{?}{\leq} \lambda \|x\| + (1-\lambda)\|y\|$$

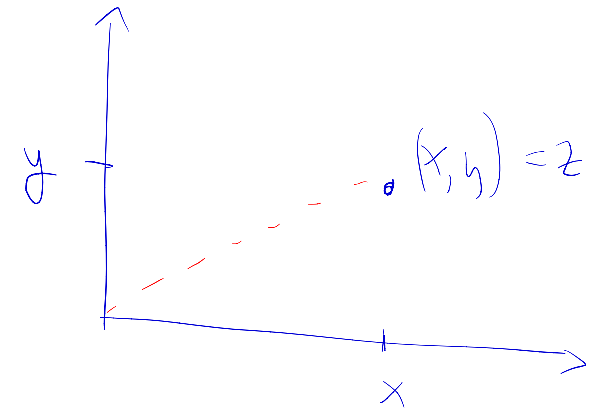
$$\|\lambda x + (1-\lambda) y\| \leq \|\lambda x\| + \|(1-\lambda) y\| \quad \left[\text{TRIANGULAR INEQ.}\right]$$

$$= \lambda \|x\| + (1-\lambda)\|y\|$$

$\square$

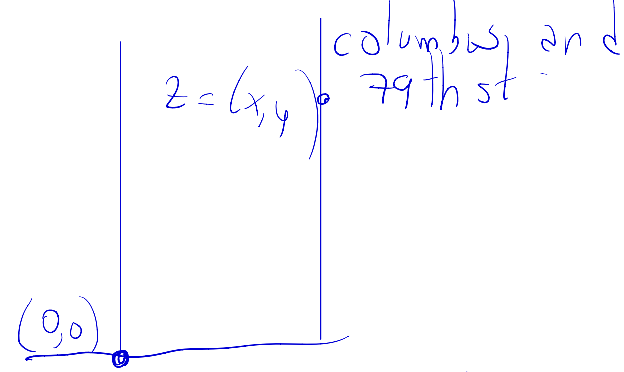
TWO EXAMPLES OF NORMS:

"USUAL" EUCLIDEAN NORM



$$\|z\|_2 = \sqrt{x^2 + y^2}$$

"TAXICAB NORM" - TO COMPUTE DISTANCE BETWEEN ROAD INTERSECTIONS IN MANHATTAN



2ND & 72nd street

$$\|z\|_1 = |x| + |y|$$

c) Let $f : \mathbb{R}^d \to \mathbb{R}$ be a convex function, and $\alpha \in \mathbb{R}$ be a number. Prove that $C = \{x : f(x) \leq \alpha\}$ is a convex set.

EXAMPLE

$f = 2x_1 + 3x_2 \quad \alpha = 7 \quad C = \{x : 2x_1 + 3x_2 \leq 7\}$

C convex if for all $x, y \in C$, we have $z = \frac{1}{2}x + \frac{1}{2}y \in C$

To show: for all $x, y \in C = \{q : f(q) \leq \alpha\}$

we have $z = \frac{1}{2}x + \frac{1}{2}y \in C = \{q : f(q) \leq \alpha\}$

$f(z) \overset{?}{\leq} \alpha$

$f(z) = f\left(\frac{1}{2}x + \frac{1}{2}y\right)$ (by def. of z)

$\leq \frac{1}{2}f(x) + \frac{1}{2}f(y)$ (by convexity of f with $\lambda = \frac{1}{2}$)

$\leq \frac{1}{2}\alpha + \frac{1}{2}\alpha$ (since $x, y \in C \Rightarrow f(x) \leq \alpha, f(y) \leq \alpha$)

$= \alpha \quad \square$

d) Let $\|x\|$ be any norm, and $\alpha \in \mathbb{R}$ be a number. Prove that $C = \{x : \|x\| \leq \alpha\}$ is a convex set.

We showed:

(I) $\|x\|$ is a convex function

(II) f convex, $\alpha \in \mathbb{R}$ $\Rightarrow$ $C = \{x : f(x) \leq \alpha\}$ is convex

(I), (II) imply the thesis

e) Prove that the portfolio optimization problem seen in class

$$
\begin{array}{ll}
\max & f(x) = 20x_1 + 16x_2 \quad -\theta[2x_1^2 + x_2^2 + (x_1+x_2)^2] \\
\text{subject to:} & \\
& \begin{aligned} x_1 + x_2 &\leq 5 \\ x_1 &\geq 0 \\ x_2 &\geq 0 \end{aligned}
\end{array}
$$

can be formulated as a convex programming problem.

flip the objective function

$$- \min \overbrace{-20x_1 - 16x_2 + \theta\left[2x_1^2 + x_2^2 + (x_1+x_2)^2\right]}^{g(x)}$$

$$\boxed{\begin{aligned} x_1 + x_2 &\leq 5 \\ x_1 &\geq 0 \\ x_2 &\geq 0 \end{aligned}}$$

To show: convex function

forms a convex set ✓ (since all constraints are linear)

To prove, we observe that $g(x)$ is twice differentiable and employ the lemma

g convex if and only if $\dfrac{\partial^2 g}{\partial^2 x_1} \geq 0$; $\dfrac{\partial^2 g}{\partial x_1^2} \cdot \dfrac{\partial^2 g}{\partial^2 x_2} - \left(\dfrac{\partial^2 g}{\partial x_1 \partial x_2}\right)^2 \geq 0$

$$g(x) = -20x_1 - 16x_2 + \Theta\left[2x_1^2 + x_2^2 + (x_1 + x_2)^2\right]$$

$$= -20x_1 - 16x_2 + \Theta\left[2x_1^2 + x_2^2 + x_1^2 + x_2^2 + 2x_1x_2\right]$$

TO BE CHECKED.

$$\frac{\partial^2 g}{\partial x_1^2} \geq 0 \; ; \quad \overbrace{\frac{\partial^2 g}{\partial x_1^2} \cdot \frac{\partial^2 g}{\partial x_2^2} - \left(\frac{\partial^2 g}{\partial x_1 \partial x_2}\right)^2}^{*} \geq 0$$

$$\frac{\partial g}{\partial x_1} = -20 + \Theta\left[4x_1 + 2x_1 + 2x_2\right]$$

$$\frac{\partial^2 g}{\partial x_1^2} = 6\Theta \overset{?}{\geq} 0$$ YES SINCE $\Theta \geq 0$ by hypothesis

$$\frac{\partial g}{\partial x_2} = -16 + \Theta\left[2x_2 + 2x_2 + 2x_1\right] ; \quad \frac{\partial^2 g}{\partial x_2^2} = 4\Theta ; \quad \frac{\partial^2 g}{\partial x_1 \partial x_2} = 2\Theta$$

$$* = 6\Theta \cdot 4\Theta - (2\Theta)^2 = 24\Theta^2 - 4\Theta^2 = 20\Theta^2 \geq 0 \qquad \square$$

Problem 3

Consider the following problem, faced by a website over the course of T times. There are n advertisers, each of which wishes to show one ad to multiple users. At time t, there are u_t users visiting the website, and the website shows to each of them exactly one of the n ads (the same add can be shown to multiple users). To each advertiser $i = 1, \dots, n$, the following quantities are associated:

- c_i: this is the minimum number of ads of advertiser i that need to be shown in total, by contract, over the T times;
- $p_{i,t}$: this is the amount an advertiser is willing to pay if their ad is shown to a user that visits the website in time t;
- b_i: this is the maximum budget advertiser i is willing to pay in total over the T times.

a) Define variables

$$x_{i,t} = \text{ number of times add } i \text{ is shown to users arriving at time } t.$$

∘ Describe the amount of money advertiser i will pay as a function of $x_{i,t}$ and other parameters of the problem.

$$\circ = \min \Big\{ \underbrace{b_i}_{\text{AN ADVERTISER CANNOT EXCEED THEIR BUDGET}}, \underbrace{\sum_{t=1}^{T} p_{i,t} x_{i,t}}_{\text{total amount an advertiser is supposed to pay for showing their ads}} \Big\}$$

b) Formulate the problem of assigning users to advertisers over the T times as to maximize the revenue of the website ($\equiv$ total expense by the advertisers). You can assume all variables to be fractional.

$$\text{MAX} \sum_{i=1}^{n} \min \left\{ b_i , \sum_{t=1}^{T} p_{i,t} x_{i,t} \right\}$$

$$\sum_{i=1}^{n} x_{i,t} \le U_t \quad \text{for } t = 1, \dots, T$$

[AT EVERY TIME t, THE TOTAL NUMBER OF ADS SHOWN CANNOT EXCEED THE NUMBER OF USERS]

$$\sum_{t=1}^{T} x_{i,t} \ge c_i \quad \text{for } i = 1, \dots, n$$

[WE HAVE TO SHOW i'S AD AT LEAST c_i TIMES]

$$x_{i,t} \ge 0$$

c) Show that the problem in b) can be formulated as a convex programming problem. You can use the following fact: if $f, g : \mathbb{R}^d \to \mathbb{R}$ are linear functions, then $h : \mathbb{R}^d \to \mathbb{R}$ is a convex function, where $h(x) = \max(f(x), g(x))$ for each $x \in \mathbb{R}^d$.

$(\colon)$ can be rewritten as

$$-\min \sum_{i=1}^{n} -\min\left\{b_i, \sum_{t=1}^{T} p_{i,t} x_{i,t}\right\}$$

$$\sum_{i=1}^{n} x_{i,t} \leq u_t \quad \text{for } t = 1, \dots, T$$

$$\sum_{t=1}^{T} x_{i,t} \geq c_i \quad \text{for } i = 1, \dots, n$$

$$x_{i,t} \geq 0$$

or equivalently

$$-\min \sum_{i=1}^{n} \max\left\{\overbrace{-b_i}^{f}, \overbrace{-\sum_{t=1}^{T} p_{i,t} x_{i,t}}^{g}\right\}$$

$$\sum_{i=1}^{n} x_{i,t} \leq u_t \quad \text{for } t = 1, \dots, T$$

$$\sum_{t=1}^{T} x_{i,t} \geq c_i \quad \text{for } i = 1, \dots, n$$

$$x_{i,t} \geq 0$$

$(\therefore)$

f, g are linear functions, thus using the hint, $\max\left\{-b_i, -\sum_{t=1}^{T} p_{i,t} x_{i,t}\right\}$ is convex. Since the sum objective functions is convex (see Problem 1), the objective function of $(\therefore)$ is convex.

Constraints are linear, thus they form a convex set

$\Rightarrow$ $(\therefore)$ is a convex optimization problem